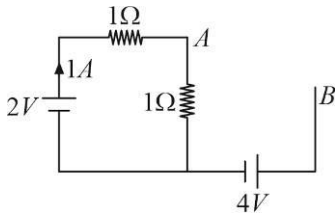


Solutions to JEE Advanced - 7 | JEE 2022 | Paper - 2

PHYSICS

1.(BD)



In steady state, the circuit reduces to

 $\Rightarrow$ By KVL,

$$\Delta V_{BA} = 3V$$

And power by 2 Volt battery = $2 \times 1 = 2W$

2.(ABC)

- (A) Let the required distance be l_1 . The lower part AP of the rod mass $\frac{m}{L}l_1$ and the upper part BP has mass $\frac{m}{L}(L-l_1)$. When heated the part AP will be moving down and friction on it (f_1) will be up the incline. The part BP will move up and friction on it will be down (say f_2).

For equilibrium of the entire rod

$$f_1 - f_2 = mg \sin \theta$$

$$\mu \frac{m}{L} l_1 g \cos \theta - \mu \frac{m}{L} (L - l_1) g \cos \theta = mg \sin \theta$$

$$\frac{\mu}{L} (l_1 - L + l_2) = \tan \theta \Rightarrow 2l_1 = L + \frac{L}{\mu} \tan \theta$$

$$l_1 = \frac{L}{2} \left[1 + \frac{\tan \theta}{\mu} \right] = 0.875L$$

- (B) In this case the friction on upper part will be up and that on the lower part will be down. Proceeding in similar way as above we can get

$$l_2 = \frac{L}{2} \left[1 - \frac{\tan \theta}{\mu} \right] = 0.125L$$

- (C) During expansion P is fixed but Q moves down. Then during contraction Q remains fixed. Therefore, in one cycle the point Q has moved down.

3.(ACD)

$$\left| \frac{\Delta Y}{Y} \right|_{\max} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2\Delta d}{d} + \frac{\Delta x}{x}$$

$$= \frac{0.1}{30} + \frac{1}{200} + \frac{2(0.001)}{0.060} + \frac{0.001}{0.200} = 0.0466; \quad \left| \frac{\Delta Y}{Y} \right|_{\max} = 4.66\%$$

4.(AD) Angular velocity of earth $\omega_1 = \frac{\pi}{12} \text{ rad/hr}$

Angular velocity of satellite $\omega_2 = \frac{\pi}{3} \text{ rad/hr}$

$$(\omega_2 - \omega_1)t = 2\pi; \quad t = 8 \text{ hrs}$$

$$T^2 \propto r^3; \quad \omega' = \frac{\pi}{24} \text{ hrs}$$

$$(\omega' - \omega_1)t = 2\pi; \quad t = 48 \text{ hrs}$$

5.(ABD) After time t , mass of plank with sand will be $m_0 + \mu t$ and let its velocity be v . In next small time interval dt , let dm mass of sand falls with speed u and sticks to it.

Change in momentum of dm :

$$d\vec{p} = dm(\vec{v}\hat{i}) - dm(-u\hat{j}) = dm(\vec{v}\hat{i} + u\hat{j})$$

$$\text{Force on } dm = \frac{d\vec{p}}{dt} = \frac{dm}{dt}(\vec{v}\hat{i} + u\hat{j}) = \mu v\hat{i} + \mu u\hat{j}$$

$$\therefore \text{Force on plank} = -\mu v\hat{i} - \mu u\hat{j} \Rightarrow F_{\text{horizontal}} \propto v \text{ and } F_{\text{vertical}} \propto u$$

$$\therefore u = \sqrt{2gh} \quad \therefore F_{\text{vertical}} \propto \sqrt{h}$$

Horizontal component of this force causes deceleration of plank.

$$\therefore (m_0 + \mu t) \frac{dv}{dt} = -\mu v \Rightarrow \int_{v_0}^v \frac{-dv}{v} = \int_0^t \frac{\mu dt}{m_0 + \mu t} \Rightarrow \ln \frac{v_0}{v} = \ln \frac{m_0 + \mu t}{m_0}$$

$$\therefore \frac{v_0}{v} = \frac{m_0 + \mu t}{m_0} \Rightarrow v = \frac{m_0 v_0}{m_0 + \mu t}$$

Let time taken to cross be T .

$$\frac{dx}{dt} = \frac{m_0 v_0}{m_0 + \mu t} \Rightarrow \int_0^\ell dx = m_0 v_0 \int_0^T \frac{dt}{m_0 + \mu t}$$

$$\Rightarrow \ell = \frac{m_0 v_0}{\mu} \ln \frac{m_0 + \mu T}{m_0} \Rightarrow m_0 + \mu T = m_0 e^{\frac{\ell \mu}{m_0 v_0}}$$

$\therefore$ Velocity after crossing

$$v = \frac{m_0 v_0}{m_0 + \mu T} = \frac{m_0 v_0}{m_0 e^{\frac{\ell \mu}{m_0 v_0}}} = v_0 e^{-\frac{\ell \mu}{m_0 v_0}}$$

$$a = \frac{dv}{dt} = \frac{-m_0 v_0 \mu}{(m_0 + \mu t)^2} = \frac{-\mu}{m_0 v_0} \left(\frac{m_0 v_0}{m_0 + \mu t} \right)^2 = \frac{-\mu}{m_0 v_0} v^2$$

6.(BC) Use conservation of momentum

$$\lambda = \frac{h}{p}$$

$$\text{Activity} = \lambda N$$

7.(B) Time after which block will move

$$F \cos \theta \geq \mu_s N$$

$$F \cos \theta \geq \mu_s (mg - F \sin \theta)$$

$$F \geq \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta)}; \quad t \geq \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta) \alpha}$$

$$t_{\min} = \frac{\mu_s mg}{\alpha \sqrt{1 + \mu_s^2}} = 2 \text{ sec}$$

When $\theta = 37^\circ$

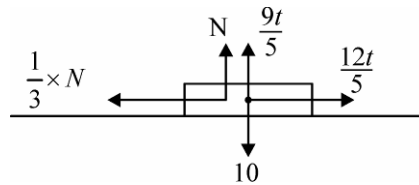
8.(C) For $\theta = 37^\circ$, after $t = 2 \text{ sec}$

$$N = 10 - \frac{9t}{5}$$

Block leaves contact when $N = 0$

$$\Rightarrow 9t/5 = 10$$

$$t = \frac{50}{9} \text{ s}$$



Impulse = $\Delta p = mv$

$$\Rightarrow v = \int_2^{50/9} \left(\frac{12t}{5} - \frac{N}{3} \right) dt = \int_2^{50/9} \left(3t - \frac{10}{3} \right) dt \approx 28 \text{ m/s}$$

9.(D)

10.(A)

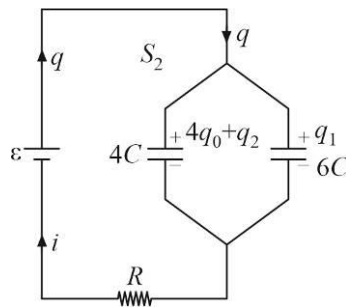
Applying Kirchoff Law

$$\frac{-q_1}{6C} - iR + \varepsilon = 0$$

$$-\left(\frac{4q_0 + q_2}{4C} \right) - iR + \varepsilon = 0$$

$$q = q_1 + q_2$$

$$i = \frac{dq}{dt}$$



On solving equation, we get results

11.(B)

$$12.(A) \quad A = \int y dx = \int_0^R \left(x \tan \theta - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 \theta} \right) dx$$

$$= \frac{2u^4}{3g^2} \sin^3 \theta \cdot \cos \theta$$

$$\text{Now, } \frac{dA}{d\theta} = 0; \quad \theta = 60^\circ$$

Since collision is perfectly inelastic hence velocity of ball after collision is only in x-direction.

1.(2) From conservation of energy

$$mg \left(\frac{a}{3} \sin \theta \right) = \frac{1}{2} (I_0) \omega^2$$

$$\omega = \sqrt{\frac{3g \sin \theta}{2a}}; \quad \Sigma F_y = ma_y$$

$$mg \cos \theta - N = m \left(\frac{a}{3} \alpha \right)$$

At slipping

$$\Sigma F_x = ma_x; \quad \mu N - mg \sin \theta = m \left(\frac{a}{3} \right) \omega^2$$

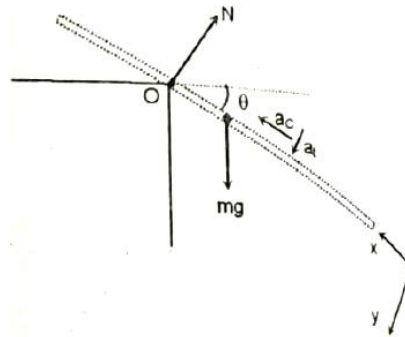
Torque equation about O

$$mg \left(\frac{a}{3} \cos \theta \right) = \left(\frac{4}{9} ma^2 \right) \alpha$$

On solving

$$N = \frac{3mg}{4} \cos \theta; \quad \mu \left(\frac{3mg}{4} \cos \theta \right) = \frac{3}{2} mg \sin \theta$$

$$\mu = 2 \tan \theta$$



2.(3) Net deviation

$$\delta = (i - r) + (\pi - 2r) + (i - r) = 2(i - r) + (\pi - 2r)$$

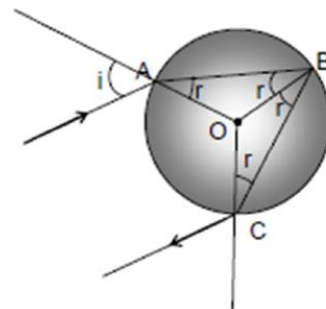
$$\text{For minimum deviation } \frac{d\delta}{di} = 0$$

$$\frac{dr}{di} = \frac{1}{2}$$

$$\text{Snell's law; } \sin i = \mu \sin r$$

$$\frac{dr}{di} = \frac{\cos i}{\mu \cos r}$$

$$\text{On solving; } \sin^2 i = \frac{4}{3} \left[1 - \frac{\mu^2}{4} \right]; \quad \sin i = \sqrt{\frac{2}{3}}$$



3.(1.20)

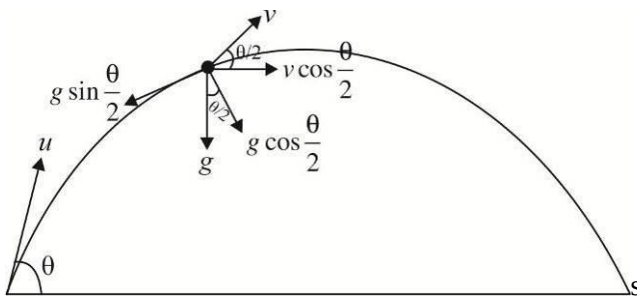
$$N = N_0 e^{-\lambda t} \text{ (where } \frac{N}{N_0} = \left(\frac{1}{2} \right)^2 = \frac{1}{4} \text{)}$$

$$N_{\text{decays}} = N_0 (1 - e^{-\lambda t}) = 10^{13} \left(1 - \frac{1}{4} \right) = \frac{3}{4} \times 10^{13}$$

$$Q = 1.6 \times 10^{-19} \times \frac{3}{4} \times 10^{13} = 1.2 \mu\text{C}$$

$$4.(2) \quad v \cos \frac{\theta}{2} = u \cos \theta \Rightarrow v = \sqrt{10} m/s.$$

$$\text{Using } a_n = \frac{v^2}{R}, R = \frac{10}{g \cos 30^\circ} \Rightarrow \sqrt{3}R = 2$$



5.(2) The field from P to C will be perpendicular to line PC . Hence, potential at P = Potential at C .

$$\frac{K(\sigma \times 2\pi R^2)}{R} = V_C$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\sigma \times 2\pi R}{R} = V_C = \frac{\sigma R}{2\epsilon_0} = 2 \text{ SI unit}$$

6.(5) From conservation of energy

$$(2M.g.3R) - (M.g.3R)$$

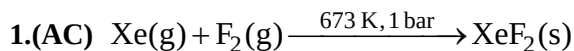
$$= \frac{1}{2} \left\{ \frac{M(6R)^2}{12} + M(3R)^2 + (2M\{3R\})^2 \right\} \omega^2$$

$$= \frac{1}{2} (30MR^2) \omega^2 = 15(MR^2) \omega^2$$

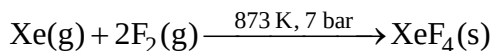
$$3MgR = 15MR^2 \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{g}{5R}}$$

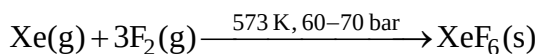
CHEMISTRY



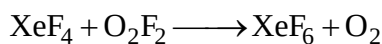
(xenon in excess)



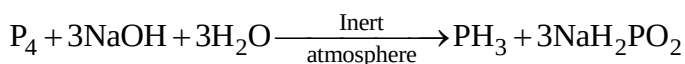
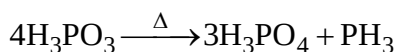
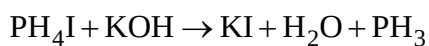
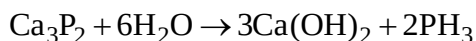
(1 : 5 ratio)



(1 : 20 ratio)



2.(ABCD)



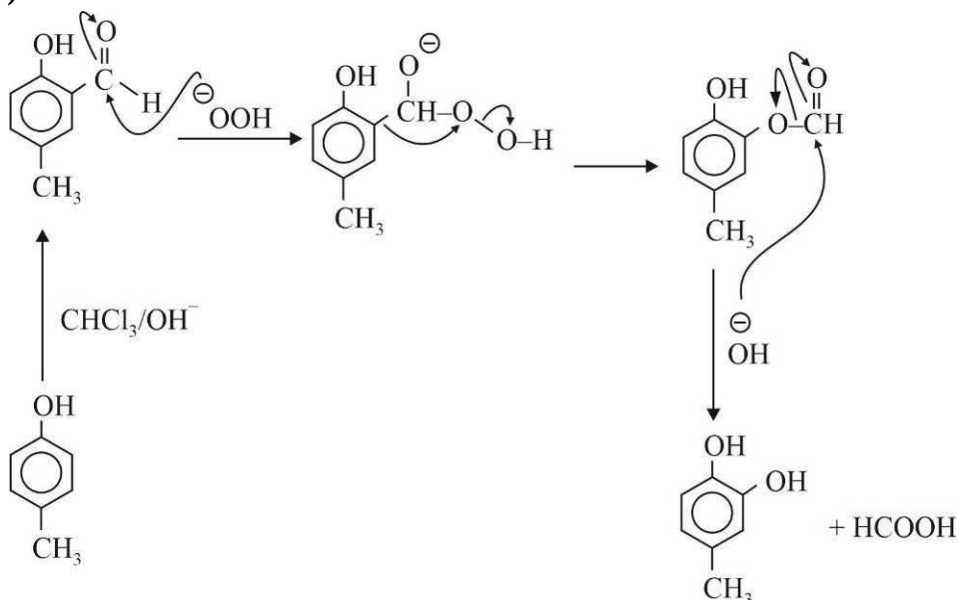
3.(B) $\text{ClF}_3 \rightarrow$ Planar distorted-T-shape and polar

$\text{XeF}_5^- \rightarrow$ Pentagonal planar and non-polar

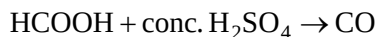
$\text{PCl}_5 \rightarrow$ Non-planar and non-polar

$\text{BrF}_5 \rightarrow$ Non-planar and polar

4.(ABD)



Hence A will be HCOOH .

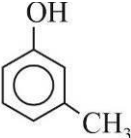


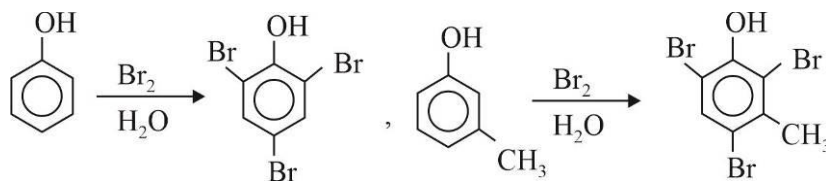
CO is a neutral gas as well as very good reducing agent.

5.(BD)

Metals having higher reduction potentials are reduced by metals having lower reduction potentials.

6.(AC)

All the ortho and para positions of  and  are vacant and activated for electrophilic substitution reactions.



7.(B) $\frac{K_1}{K_2} = \frac{1}{2} = \frac{[B]}{[C]} = \frac{x}{y}$

$t = 0$ A_0 = No. of moles of A at $t = 0$

$t = t_0$ $A_0 - x - y$ = No. of moles of A at $t = t_0$

$$x + y = \frac{A_0}{2} = \frac{2}{2} = 1 \quad \dots(1)$$

$$\frac{x}{y} = \frac{1}{2} \Rightarrow 2x = y$$

Putting value of x in equation (1)

$$x = \frac{1}{3} \text{ \& } y = \frac{2}{3}$$

$$n_T = n_A + n_B + n_C = 1 + \frac{2}{3} + \frac{4}{3} = \frac{9}{3} = 3$$

8.(B) $\frac{n_B}{n_C} = \frac{K_1}{K_2}$ at any time

9.(C) $\Delta T_b = iK_b \cdot m$

$$6 = i \times 2 \times 1$$

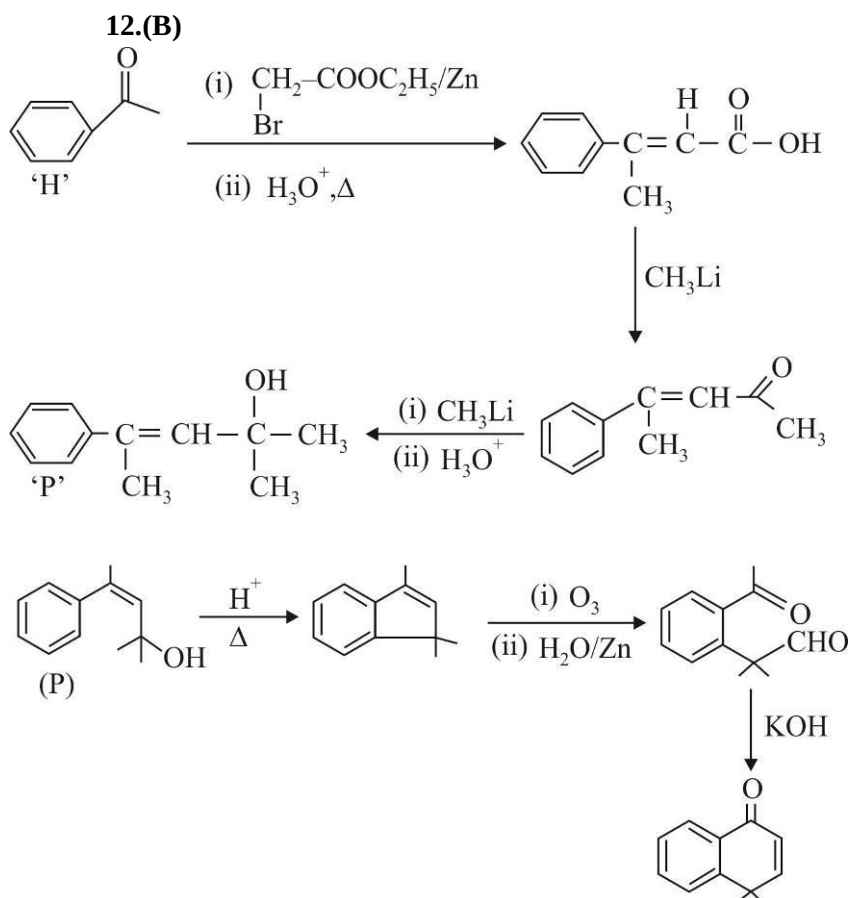
$$i = 3$$

Complex will dissociate in three fragments only so complex is $[\text{Co}(\text{NH}_3)_5 \text{Cl}] \text{Cl}_2$

10.(A) Moles of $\text{AgCl} = 1$

$$\text{Moles of AgCl/mole of complex} = \text{ionizable } \text{Cl}^- \text{ in complex} = 1 \quad [\text{Co}(\text{NH}_3)_4 \text{Cl}_2] \text{Cl}$$

11. (C)



1. (0) Benzaldehyde does not reduce Fehling solution.

2. (6) Molecular weight of decapeptide = 796 g/mole
 Total bonds to be hydrolysed = $(10 - 1) = 9$ per molecules
 Total weight of H_2O added = $9 \times 18 = 162$ g/mol.
 Total weight of hydrolysis product = $796 + 162 = 958$
 Total weight % of glycine = 47%
 Total weight of glycine in product = $\frac{796 \times 47}{100} \approx 450$ gm
 Molecular wt of glycine in product = 75 g/mol.
 $\therefore$ no. of glycine units = $450/75 = 6$.

3. (8) I_3^+ , XeF_4 , C_2F_4
 SO_3 , NOCl , ClF_3
 F_2CO , XeF_5^-

4. (6) $\text{—}\overset{\text{O}}{\parallel}\text{C—OEt} \rightarrow 2$ moles, $\text{—}\overset{\text{O}}{\parallel}\text{C—H} \rightarrow 1$ mole
 $\text{—OH} \rightarrow 1$ mole, $\text{—SH} \rightarrow 1$ mole
 Epoxide -1 mole so total 6 moles.

5.(8) NiS and $\text{CoS} \rightarrow$ black in colour.

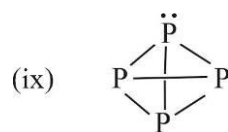
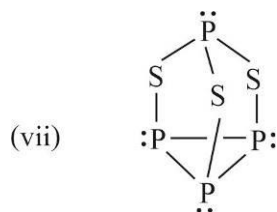
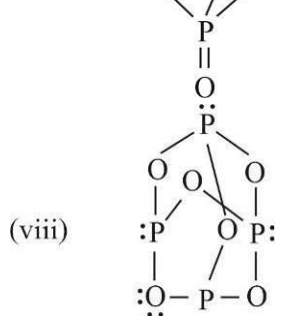
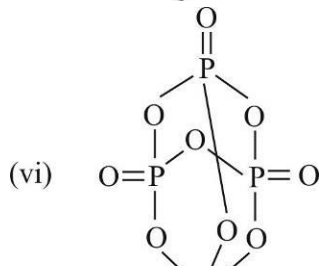
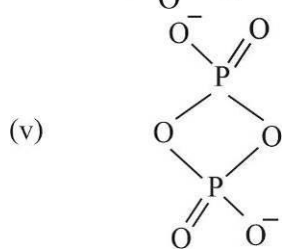
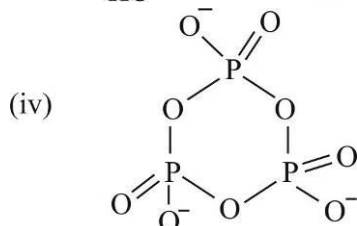
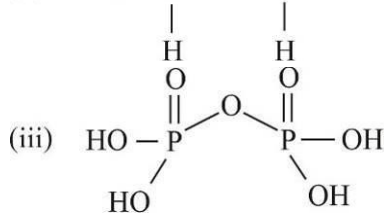
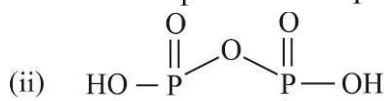
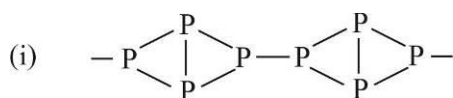
$$A = 2$$

$$B = 4$$

$$C = 0$$

$$D = 2$$

6.(3)



MATHEMATICS

1.(BC) $A \cdot \text{adj}(2B) = 16I$

$A(4 \text{ adj}B) = 16I$

$\Rightarrow A \cdot \text{adj}B = 4I \quad \Rightarrow A|B| \cdot B^{-1} = 4I$

$B \cdot \text{adj}A = B \cdot \text{adj}(2B) = 4B \cdot \text{adj}B = 4|B|I_3 = 8I_3$

$A \cdot \text{adj}B = 4I$

$\Rightarrow A^{-1}A \cdot \text{adj}B \quad \Rightarrow \text{adj}B = 4I$

$\Rightarrow A^{-1} \cdot A \cdot \text{adj}B = 4A^{-1} \quad \Rightarrow \text{adj}B = 4A^{-1}$

$A^{-1} \cdot \text{adj}B = A^{-1} \cdot (4A^{-1}) = 4(A^{-1})^2$

$(A^{-1}(\text{adj}B))^{-1} = (4A^{-1}A^{-1})^{-1} = \frac{1}{4}A^2 = \frac{1}{4}(2B)^2 = B^2$

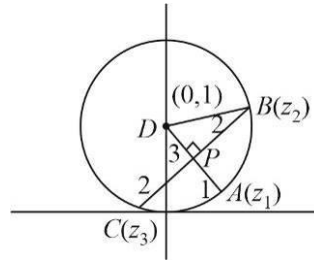
2.(AC) $\frac{3z_1 + i}{4} = \frac{2z_2 + 2z_3}{4}$

$AD = 1; \quad DP = \frac{3}{4}$

$r^2 = BP^2 + DP^2 \Rightarrow BP = 1 - \frac{9}{16} = \frac{\sqrt{7}}{4}$

$BC = \frac{\sqrt{7}}{2}$

Area of quad $ABCD = \frac{1}{2} \times \frac{\sqrt{7}}{2} \times 1 = \frac{\sqrt{7}}{4}$



3.(AC) Let $S = \sum_{k=1}^{\infty} \frac{P_k}{3^k} = \frac{P_1}{3} + \frac{P_2}{3^2} + \frac{P_3}{3^3} + \frac{P_4}{3^4} + \frac{P_5}{3^5}$

$S = \frac{1}{3} + \frac{1}{3^2} + \frac{2}{3^3} + \frac{3}{3^4} + \frac{5}{3^5} + \frac{8}{3^6} + \dots + \frac{P_n}{3^n}$

$\frac{S}{3} = \frac{1}{3^2} + \frac{1}{3^3} + \frac{2}{3^4} + \frac{3}{3^5} + \dots + \frac{P_{n-1}}{3^n} + \dots$

$\frac{S}{9} = \frac{1}{3^3} + \frac{1}{3^4} + \frac{2}{3^5} + \dots + \frac{P_{n-2}}{3^n}; \quad S = \left(\frac{S}{3} + \frac{S}{9} \right) + \frac{1}{3} \Rightarrow S = \frac{3}{5}$

4.(BCD)

Let H = event that married man watches the show

W = event that married women watches the show

$P(H) = 0.4, P(W) = 0.5, P\left(\frac{H}{W}\right) = 0.7$

$P\left(\frac{W}{H}\right) = \frac{0.35}{0.4} = \frac{7}{8}, P(H \cup W) = 0.55$

5.(ABD)

$P(x)$ is an even function

$$\therefore P(x) = ax^4 + bx^2 + 1 \text{ and } P'(x) = 4ax^3 + 2bx = 2x(2ax^2 + b)$$

It has two minima. Hence, $a > 0$ and $b < 0$

So, at $x = \pm\sqrt{\frac{-b}{2a}}$, $P(x)$ has minima

$$2\sqrt{\frac{-b}{2a}} = 2 \Rightarrow b = -2a \quad \therefore \text{Maximum at } (0, 1)$$

$$\text{Also, } \frac{8\sqrt{2}}{15} = 2 \int_0^{\sqrt{\frac{-b}{a}}} (1 - (ax^4 + bx^2 + 1)) dx \Rightarrow b = -1, a = \frac{1}{2}$$

$$\text{Now, } \lim_{x \rightarrow 0} \frac{P(x) - (g(x) + g(-x))}{x^2} \Rightarrow C = \frac{1}{2}, B = -1$$

Also, $y = 1$ is tangent to $Ax^2 - x + \frac{1}{2} = f(x)$

$$\Rightarrow Ax^2 - x + \frac{1}{2} = 1 \text{ has equal roots} \Rightarrow A = -\frac{1}{2}$$

6.(ABC)

Let the roots of $az^3 + bz^2 + cz + d = 0$, $z_1 = x_1$, $z_2, z_3 = x_2 \pm iy_2$

$$\Rightarrow z_1 + z_2 + z_3 = -\frac{b}{a} \Rightarrow ab > 0$$

$$\text{Also, } z_1 z_2 z_3 = x_1(x_2^2 + y_2^2) = -\frac{d}{a} \Rightarrow ad > 0$$

$$7.(C) \quad V_n = \sum_{r=1}^n S_r = 2 \sum_{r=1}^n r = n(n+1) \quad \therefore \lim_{n \rightarrow \infty} \frac{V_n}{n^2} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$$

$$8.(B) \quad (I) \quad V_n = n(n+1) \Rightarrow V_1 < V_2 < V_3 < \dots$$

$$(II) \quad Q_n = \sum_{r=1}^n \frac{1}{S_r S_{r+1}} = \frac{1}{4} \sum_{r=1}^n \frac{1}{r(r+1)} = \frac{1}{4} \sum_{r=1}^n \left(\frac{1}{r} - \frac{1}{r+1} \right) = \frac{1}{4} \left(1 - \frac{1}{n+1} \right)$$

$$\Rightarrow Q_1 < Q_2 < Q_3 < \dots$$

$$\Rightarrow \text{Statement -1 is false}$$

9.(B) We have,

$$f(x) = \begin{cases} 2a^{-x}, & x < 0 \\ a^{-x} + a^x, & x \geq 0 \end{cases}$$

Clearly, $f(x)$ is continuous in R ,

$$\text{For } x > 0, f'(x) = -a^{-x} \log_e a + a^x \log_e a = (a^x - a^{-x}) \log_e a = \left(\frac{a^{2x} - 1}{a^x} \right) \log_e a > 0 \text{ for } x > 0$$

$\Rightarrow f(x)$ is monotonic increasing for $x > 0$

For $x < 0$, $f(x) = 2a^{-x}$

$\Rightarrow f'(x) = -2a^{-x} \log_e a < 0 \Rightarrow f(x)$ is monotonic decreasing for $x < 0$

$$\begin{aligned} 10.(B) \text{ Required area} &= \int_{-1}^0 2a^{-x} dx + \int_0^1 (a^{-x} + a^x) dx = \left\{ \frac{-2a^{-x}}{\log_e a} \right\}_{-1}^0 + \left\{ \frac{-a^{-x} + a^x}{\log_e a} \right\}_0^1 \\ &= \frac{1}{\log_e a} [-2 - (-2a)] + \frac{1}{\log_e a} \left[\left(\frac{-1}{a} + a \right) - 0 \right] = \frac{3a^2 - 2a - 1}{a \log_e a} \end{aligned}$$

11.(A) if $y = x$ in $f(x) f(y) = f(xy)$

$$\Rightarrow f(x^2) = f^2(x) \geq 0$$

Put $x = t, y = z$

$$[f(x) + f(y)]^2 = f(x^2 + y^2) \Rightarrow f(x^2 + y^2) = f^2(x) + f^2(y) + 2f(x)f(y) \geq f^2(t)$$

$$\Rightarrow f(x^2 + y^2) \geq f(x^2) \Rightarrow f \text{ is non decreasing for positive } x.$$

12.(C) Put $y = z = t = 1$

$$\Rightarrow 2(f(x) + 1) = f(x-1) + f(x+1)$$

$$f(2) = 4, f(3) = 9, f(1) = 1, f(0) = 0$$

$$f(n) = n^2 \text{ (Possible function), if } f(n-1) = (n-1)^2$$

$$2[f(n-1) + 1] = f(n-2) + f(n) \Rightarrow f(n) = n^2$$

Now, for $x = \frac{p}{q}$ (rational number)

$$f\left(\frac{p}{q}\right) f(q^2) = f(pq) \Rightarrow f\left(\frac{p}{q}\right) q^2 = p^2 q^2 \Rightarrow f\left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)^2 \text{ (True for rational number)}$$

Now, if $x \in \mathbb{R}$, let's prove for positive x since if it is proved the function is even and will follow for negative x

Assume for $x > 0$, $f(x) < x^2$

So, now a rational number ' r ' between $\sqrt{f(x)}$ and $x \Rightarrow \sqrt{f(x)} < f < x$

$$\Rightarrow f(x) < r^2 < x^2$$

$[f(r) = r^2, f \text{ is non decreasing}]$

$$\Rightarrow f(r) = r^2 \leq f(x) \text{ [contradiction]} \Rightarrow f(x) < x^2 \text{ (impossible)}$$

Similarly we can prove contradiction $f(x) > x^2$

So, only possibility $f(x) = x^2$ substituting $f(x) = x^2$ we get

$$(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$$

$$(x^2 + z^2)(y^2 + t^2) = (xy - zt)^2 + (xt + yz)^2 \quad [\text{Lagrange identity}]$$

So, $f(x) = \frac{1}{2}, f(x) = 0, f(x) = x^2$ are the required solutions

1.(8) Clearly f is a linear function.

2.(3) Let $t = x^3$ in I_2

$$\text{We have } I_2 = \frac{1}{3} \int_0^1 \frac{1}{e^t(2-t)} dt = \frac{1}{3} \int_0^1 \frac{1}{e^{1-t}(1+t)} dt \quad \Rightarrow \quad I_2 = \frac{1}{3e} \int_0^1 \frac{e^t}{(1+t)} dt = \frac{1}{3e} I_1 \Rightarrow \frac{I_1}{eI_2} = 3$$

3.(0.80)

$f(x) = (f(|x|))$ is non differentiable at 5 points.

$$\text{It means } \sum_{i=1}^5 |x_i| = 8$$

$$2|x_1| + 2|x_2| = 8; \quad x_1 + x_2 = 4$$

$$f(x) = x^2 - 4x + c; \quad \frac{1}{5} \lim_{x \rightarrow \infty} \frac{x^2(x^2 - 4x + c)}{x} = \frac{4}{5}$$

$$4.(9) \quad A = \{1, \sqrt{2}, \sqrt{3}, 2\}, B = \left\{ \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4} \right\}$$

$$m = 3^4 - ({}^3C_1 2^4 - {}^3C_2) = 36 = 2^2 3^2 \text{ hence number of divisors of } m \text{ is } 9.$$

$$5.(8) \quad \sum_{r=0}^9 {}^{20}C_{2r} \cdot {}^{20}C_{2r+2} = \frac{{}^{40}C_{22} - {}^{20}C_{11}}{2} = \frac{10}{11} ({}^{39}C_{21} - {}^{19}C_{10})$$

$$\Rightarrow a = 10 \Rightarrow xy = 40 \text{ has total order pair } (x, y) \text{ solution} = 4 \times 2 = 8$$

6.(540)

$$\text{Total matrices} = \frac{9!}{5!2!2!} = 756$$

$$B \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \text{Trivial sol. } (D \neq 0)$$

$$a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_2 b_1 c_3 - a_3 b_2 c_1$$

$$|B| = 1 \Rightarrow {}^3C_2 \times {}^6C_1 \times \frac{6!}{5!} = 36 \times 3 = 108$$

(one term = 1) (let say $a_1 b_2 c_3 = 1$) (3 ways possible)

Similarly $|B| = -1$

Total cases = 108

Number of possible for non-trivial sol. = $756 - 216 = 540$